9 Cross-tabulating two qualitative variables

Sheet 49 Thu 11 Mar. 2027 *Cross-tabulating two qualitative variables*

1. Warm-up · Quick [PRO] Express $\frac{24}{60}$ as a percentage.
   Ans. 40%.
2. Core · Quick [LEC] In a class of 30 pupils, 18 are girls. What is the relative frequency of girls? of boys?
   Ans. $\frac{18}{30} = 60\%$; 40%.
3. Core + · Classify [STA] Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.
   Ans. Nominal qualitative; ordinal qualitative (natural order); quantitative.
4. Review · Problem [MOD] A rectangle of perimeter 20 cm has length x. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.
   Ans. Width $10 - x$; $A(x) = x(10 - x)$; $A(3) = 21$, $A(5) = 25$ cm^2.
5. Thread · Quick [SOL] Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.
   Ans. $V = \pi \times 4 \times 5 = 20\pi$ cm^3.

Sheet 50 Mon 15 Mar. 2027 *Qualitative variables; two-way frequency table*

1. Warm-up · Quick [CAL] Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.
   Ans. 18; 18.
2. Core · Table [LEC] Total 40 pupils. Girls: 22; canteen users (C): 25; girls who use the canteen: 15. Complete the two-way table.

	C	Not C	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

   Ans. Girls not using the canteen 7; boys using the canteen 10; boys not using the canteen 8; total boys 18; total not using the canteen 15 (check: $7 + 8 = 15$).
3. Core + · True/false [LEC] True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.
   Ans. False. There is a natural order: it is an ordinal qualitative variable.
4. Review · Problem [CMP] Discount A: -30% on 80,000 Ar. Discount B: -20% then -10%. Which is the better deal?
   Ans. A: 56,000 Ar; B: $80{,}000 \times 0.8 \times 0.9 = 57{,}600$ Ar. A is the better deal.
5. Thread · Method [DEM] We know that $AB^2 + AC^2 = BC^2$. What conclusion can be drawn? Which property is used?
   Ans. ABC is right-angled at A: converse of Pythagoras' theorem.

Sheet 51 Tue 16 Mar. 2027 *Marginal frequencies*

1. Warm-up · Quick [PRO] Express $\frac{25}{40}$ as a percentage.
   Ans. 62.5%.
2. Core · Method [LEC] Using the table from sheet 50: calculate the marginal frequency of girls and that of canteen users.
   Ans. Girls: $\frac{22}{40} = 55\%$; canteen: $\frac{25}{40} = 62.5\%$ (marginal counts over the total count).
   Marginal frequency = row (or column) total / overall total.

3. Core + · Quick [LEC] The marginal frequency of girls is 55%. What is that of boys?
   Ans. 45%: the marginal frequencies of a variable add up to 100%.
4. Review · Quick [SUB] Calculate $(2a-b)^2$ for $a=3$ and $b=-2$.
   Ans. $(6+2)^2=64$.
5. Thread · Quick [REP] $A(-1,2)$ and $B(3,2)$. What can be said about the line (AB)? Give AB.
   Ans. Horizontal (same ordinate), parallel to the horizontal axis; $AB=3-(-1)=4$.

Sheet 52 Thu 18 Mar. 2027 *Conditional frequencies*

1. Warm-up · Quick [PRO] Give $\dfrac{15}{22}$ as a percentage rounded to the nearest tenth.
   Ans. 68.2%.
2. Core · Method [LEC] With the same table: among the girls, what is the frequency of canteen users? Among the boys?
   Ans. Girls: $\dfrac{15}{22}\approx 68.2\%$; boys: $\dfrac{10}{18}\approx 55.6\%$ (divide by the total of the subpopulation).
3. Core + · Distinguish [LEC] Distinguish between "the frequency of girls among canteen users" and "the frequency of canteen users among girls".
   Ans. $\dfrac{15}{25}=60\%$ and $\dfrac{15}{22}\approx 68.2\%$: the reference total is not the same.
4. Review · Construction [CON] Draw $[AB]$ of length 6 cm, then the circles with centres A and B and radius 4 cm, which meet at C and D. What type of quadrilateral is $ACBD$?
   Ans. A rhombus: $AC=CB=BD=DA=4$ cm (four equal sides).
5. Thread · Quick [DIV] Is 91 prime?
   Ans. No: $91=7\times 13$.
   Pitfall: it is divisible by none of 2, 3, 5.

Sheet 53 Mon 22 Mar. 2027 *Completing a two-way table*

1. Warm-up · Quick [PRO] Calculate 35% of 80.
   Ans. 28.
2. Core · Method [LEC] 200 customers: 60% are women; 25% of the women and 50% of the men buy online. Complete the two-way table.
   Ans. Women 120, of whom 30 online; men 80, of whom 40 online; online in total: 70.
   To go from conditional frequencies to counts, multiply by the size of the subpopulation.
3. Core + · Check [LEC] A table states "row total: $45+55=100$" but "column total: $60+30=90$". What can be concluded?
   Ans. The table is inconsistent: the grand total must be the same by rows and by columns; there is a data-entry or calculation error.
4. Review · True/false [PRB] True or false? Justify. After three consecutive "heads", the probability of getting "tails" on the next toss is greater than $\dfrac{1}{2}$.
   Ans. False. The tosses are independent: the probability remains $\dfrac{1}{2}$.
5. Thread · Quick [GEO] ABC is right-angled at A, $BC=10$ cm, $\widehat{ABC}=30°$. Calculate AC.
   Ans. $AC=BC\times\sin 30°=10\times 0.5=5$ cm.
   Use the sine: opposite side / hypotenuse.

Sheet 54 Tue 23 Mar. 2027 *Real data: processing a file and filtering (AND, OR, NOT)*

1. Warm-up · Quick [LEC] In a file of 1,000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy both A and B. How many satisfy "A AND B"? "A OR B"?

   Ans. 100; $400 + 300 - 100 = 600$ (the intersection must not be counted twice).

2. Core · Method [LEC] With the same data: how many satisfy "NOT A"? "A AND NOT B"?

   Ans. $1{,}000 - 400 = 600$; $400 - 100 = 300$.

3. Core + · Reading code [ALG] What is the final value of `compte`?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```

   Ans. 3 (the ages 16, 17, 16).

4. Review · Quick [NUM] Give the decimal form of $\frac{7}{8}$, then the fraction in lowest terms equal to 0.12.

   Ans. 0.875; $\frac{12}{100} = \frac{3}{25}$.

5. Thread · Formula → table [REG] $f(x) = x^2 - 2x$. Complete a table of values for $x \in \{0, 1, 2, 3\}$.

   Ans. 0 ; -1 ; 0 ; 3.

Sheet 55 Thu 25 Mar. 2027 *Critical reading of statements from the media*

1. Warm-up · Quick [PRO] A 10% increase on 40: what is the final value?

   Ans. $40 \times 1.10 = 44$.

2. Core · Critical thinking [LEC] An article claims: "60% of accidents involve sober drivers". Can we conclude that driving sober is dangerous?

   Ans. No: the reference population is missing. If the vast majority of drivers are sober, they can be the majority among those involved in accidents even if their accident risk is lower. One would have to compare accident frequencies among sober drivers and among the others.

3. Core + · True/false [LEC] True or false? Justify. "Two smokers out of three are men" means that two men out of three smoke.

   Ans. False. This confuses $P_{\text{smoker}}(\text{man})$ and $P_{\text{man}}(\text{smoker})$: the reference populations differ.

4. Review · True/false [STA] True or false? Justify. The median of a data set is always equal to its mean.

   Ans. False. Counter-example: 1, 2, 9: mean 4, median 2.

5. Thread · Reading code [ALG] What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

   Ans. $s = 1 + 2 + 3 + 4 = 10$.

   Pitfall: `range(1, 5)` stops at 4; 5 is excluded.